DATASET PROPOSAL & CREDIBLE RESOURCES DOSSIER

HACKINSIGHT: DATA-TO-DECISION (TECHFEST '26) — AIR QUALITY TRACK

Team: DataMinors	**Target Horizon:** 24-Hour Ahead Hourly PM2.5 (µg/m³)
Members: Laiba Akram, Ali Hasan Farooqui, Areeba, Bushra Abid	**Target Domain:** Lahore & Karachi Metropolitan Agglomerations
Repository: github.com/Ali-HF/HackInsight	**Warehouse Volume:** 176,392 Verified Records (DuckDB/Parquet)

1. Problem Statement & Primary Decision-Maker Context

The Operational Challenge: The megacities of Lahore and Karachi face acute, chronic air pollution emergencies. During the winter smog season (October to February), Lahore frequently registers the highest PM2.5 concentrations worldwide (>500 µg/m³), imposing severe public health crises, widespread pediatric respiratory morbidity, and multi-billion-rupee economic shutdowns. Conversely, government reference monitors in Pakistan are sparse, geographically restricted, and suffer from recurrent data downtime.

Primary User & Decision Driven: The primary user is the *Provincial Smog Response Officer* at the Punjab and Sindh Environmental Protection Departments (EPD / EPA). Every morning, this officer must enact statutory emergency actions: issuing district-specific public health advisories, directing school closures, and ordering targeted industrial or vehicular restrictions 24 hours in advance. Our system provides this officer with calibrated, hourly 24-hour-ahead forecasts equipped with conformal 80% uncertainty bounds and grounded citations.

2. Credible Resource Inventory & Provenance Verification

In strict compliance with HackInsight Chapter 4 data governance mandates, all model training and evaluation rely on at least two independent, credible, and fully licensed sources, including an authoritative off-track resource with citable peer-reviewed provenance.

Source 1 (Primary Ground Network — Off-Track Credible Resource): PAQI via NASA ASDC DAAC

- **Dataset Name:** Low-Cost Sensor AQ: Pakistan Air Quality Initiative (PAQI)
- **Originating Provider:** Pakistan Air Quality Initiative (Karachi / Lahore, Network PI: Abid Omar)
- **Archive Publisher & DOI:** NASA Langley Atmospheric Science Data Center (ASDC) DAAC — **DOI: 10.5067/LOW-COST-SENSORS-AQ/PAKAIRQUALITY**
- **Off-Track Justification & Scientific Credibility:** Hosted by NASA ASDC under the NASA Low-Cost Air Quality Sensor Harmonization project. The dataset has undergone scientific vetting and peer-reviewed analysis presented at the European Geosciences Union (EGU General Assembly 2024, *Omar & Rosales, doi:10.5194/egusphere-egu24-453*). Fulfills the 'credible off-track source' requirement.
- **Coverage & Volume:** 79,645 hourly timestamps spanning October 2016 through March 2026 across 210 sensor channels nationwide. Karachi contains 18 stations with 143,382 valid PM2.5 readings; Lahore contains 68 stations with 479,145 valid PM2.5 readings.
- **Format & Checksum:** ICARTT standardized atmospheric format. Verified SHA-256: 892dc29a5d38af1e9f7a89996592faad0b2d93cb66bf3ebf72406d748819fe13 (181.9 MB).

Source 2 (Independent Physical Meteorological Covariates): NASA POWER Project

- **Dataset Name:** NASA POWER Hourly Point Reanalysis (Prediction Of Worldwide Energy Resources)
- **Publisher & Host:** NASA Langley Research Center / Global Modeling and Assimilation Office (GMAO)
- **Independence:** Satellite observations and assimilation physics (MERRA-2, GEOS-IT, CERES) completely physically independent of ground sensor hardware.
- **Covariates Ingested:** Hourly 2m Temperature (T2M), Relative Humidity (RH2M), 10m Wind Speed (WS10M), Wind Direction (WD10M), Surface Pressure (PS), Precipitation (PRECTOT), and Dew Point (T2MDEW). Latency: 2–3 days.
- **Licensing & Access:** Open Access / CC BY 4.0. No restrictions on scientific modeling and decision-support deployment.
- **Volume:** 6 localized JSON payloads representing 26,304 hourly atmospheric measurements per city (2022–2024).

Source 3 (Real-Time Ingestion & Out-of-Network Generalization): OpenAQ v3 API

- **Aggregator:** OpenAQ Foundation (openaq.org), Global Environmental Data Commons.
- **Deployment Role:** Ingests live hourly observations from independent ground sensors (e.g., Hawanama network at Dawlance Karachi, US Consulate reference monitors) to drive hourly Prefect orchestration flows and evaluate out-of-network model generalization.
- **License:** CC BY 4.0 / Public Domain open sensor networks.

Source 4 (Statutory & Regulatory Standards): Pak-EPA NEQS & WHO 2021

- **Pakistan Statutory Standard:** Pakistan Environmental Protection Agency National Environmental Quality Standards (Pak-EPA NEQS 2010), statutory 24-hour PM2.5 limit of **35.0 µg/m³**.
- **Global Health Guideline:** World Health Organization 2021 Air Quality Guidelines, 24-hour PM2.5 threshold of **15.0 µg/m³**.
- **Role in Pipeline:** Injected into the Insight Agent (RAG) knowledge engine to guarantee that all alert classifications, action recommendations, and school advisories are legally and scientifically grounded.

3. Manifest Register & Cryptographic Hash Verification

In conformance with HackInsight Chapter 4, every raw and processed artifact ingested into the warehouse is recorded in data/manifests/manifest.csv with UTC fetch timestamp, file size, license terms, and SHA-256 hash:

Source ID	Resource & Publisher	Format	Size	SHA-256 Checksum (Hex)	License
SRC-PAQI-PM25	PAQI Network / NASA ASDC DAAC	ICARTT	181.9 MB	892dc29a5d38af1e9f7a8999...fe13	NASA ASDC Open
SRC-POWER-LHE-22	NASA POWER Lahore 2022 (Hourly Met)	JSON	1.13 MB	c44418088e3d2549014b7b8d...0c98	CC BY 4.0
SRC-POWER-LHE-23	NASA POWER Lahore 2023 (Hourly Met)	JSON	1.13 MB	ec27c89ad6b6210a797f2480...3e42	CC BY 4.0
SRC-POWER-LHE-24	NASA POWER Lahore 2024 (Hourly Met)	JSON	1.13 MB	859c956e5b5fa7bfb560d8ec...4d59	CC BY 4.0
SRC-POWER-KHI-22	NASA POWER Karachi 2022 (Hourly Met)	JSON	1.14 MB	56023a417c3e956f59300efe...dad9	CC BY 4.0
SRC-POWER-KHI-23	NASA POWER Karachi 2023 (Hourly Met)	JSON	1.14 MB	cbf5e4a807ccfd279c02d187...6c93	CC BY 4.0
SRC-POWER-KHI-24	NASA POWER Karachi 2024 (Hourly Met)	JSON	1.14 MB	93443e22f9807ff8cfa6fec9...91df5	CC BY 4.0

4. Three-Tier Warehouse Architecture & Pipeline Flow

Our data infrastructure is structured into three clean, audit-traced storage layers:

1. **Raw Immutable Layer (data/raw/):** Read-only ICARTT and JSON files with strict hash locks. Never modified in-place.
2. **Cleaned Interim Layer (data/interim/):** Station-parsed tabular frames with standardized UTC ISO 8601 timestamps, explicit units (µg/m³, °C, %, m/s, hPa), -9999 nodata masking, and string whitespace normalisation.
3. **Processed Gold Warehouse (data/processed/gold_pm25_forecasting.parquet & data/warehouse.duckdb):** Strict 16-column analytical table containing 176,392 hourly records across 30 verified stations. Features stable cryptographic row IDs, causal hourly forward-fill (max 2h cap), zero future leakage, and time-partitioned indexing.

5. Autonomous Quality Governance: DataQualityAgent

Unlike brittle manual data cleaning, data ingestion is governed by an autonomous **Data Quality Agent** executing deterministic policies:

- **FIX (Safe Deterministic Repair):** Strips whitespace, standardizes station naming, formats ISO timestamps. Logged as FIX.
- **FILL (Controlled Bound Imputation):** Forward-fills short meteorological gaps strictly up to 2 hours ($gap \le 2h$). All imputed cells are marked. Target PM2.5 is **never** imputed to prevent ground truth contamination.
- **FLAG (Anomaly Isolation without Deletion):** Detects sensor flatline episodes ($\ge 6h$ identical float values), temporal gaps ($>1h$), and physical range extremes ($>800 µg/m³$). Flagged records are preserved in audit logs (reports/data_quality/quality_log.csv) with complete rationale for expert inspection.

6. Compliance & Track Selection Sign-Off

The datasets proposed and utilized in HackInsight provide deep historical volume (176k+ rows, 8+ years of telemetry), provable provenance from world-class space agencies (NASA ASDC DAAC and NASA POWER), independence between target and covariates, and legal compliance for public and scientific decision-support. All assets are fully reproducible via make data.

Submitted By: Team DataMinors	**Track Verified:** Air Quality (Pakistan)
Date: October 21, 2026	**Integrity Status:** 100% Verified (SHA-256 Validated)